Maincrafts Internship Task 1

Linear Regression Model for House Price Prediction

Artificial Intelligence & Machine Learning – Task 1

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1. Introduction

The objective of this project is to build and evaluate a **Linear Regression model** to predict house prices using the California Housing dataset. This task demonstrates the complete machine learning workflow, including data loading, exploratory data analysis (EDA), preprocessing, model training, evaluation, and reporting.

Machine Learning regression models are widely used in real-world applications such as real estate price prediction, business forecasting, and financial analysis. Linear Regression was chosen because it is simple, interpretable, and effective for understanding relationships between variables.

2. Dataset Description

The project uses the **California Housing Dataset**, available in the **scikit-learn** library.

The dataset contains information collected from California housing districts.

Features Used

- **MedInc** – Median income of residents
- **HouseAge** – Average age of houses
- **AveRooms** – Average number of rooms
- **AveBedrms** – Average bedrooms
- **Population** – Population in block group
- **AveOccup** – Average occupancy
- **Latitude & Longitude** – Geographic location

Target Variable

- **MedHouseVal** – Median house value (price)

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the dataset structure and relationships between features.

Key Observations

- Median income shows the strongest positive correlation with house price.
- Geographic location significantly influences housing values.
- Most features follow reasonable distributions without severe anomalies.
- No missing values were found in the dataset.

Visualizations Used

- Correlation Heatmap
- Feature Distribution Histograms
- Scatter Plot (Income vs House Price)

EDA helps identify patterns and improves model understanding before training.

4. Data Preprocessing

Before training the model, the following preprocessing steps were performed:

- Dataset converted into a pandas DataFrame.
- Features and target variable separated.
- Dataset split into **training (80%)** and **testing (20%)** sets using Train-Test Split.

This ensures unbiased model evaluation.

5. Model Development

A **Linear Regression** algorithm from scikit-learn was used.

Linear Regression models the relationship between independent variables (features) and dependent variable (house price) using a linear equation:

$$y = b_0 + b_1x_1 + b_2x_2 + \ldots + b_nx_n$$

Steps Followed

1. Initialize LinearRegression model

2. Train model using training data
3. Predict house prices on test data

6. Model Evaluation

Model performance was evaluated using standard regression metrics.

Evaluation Metrics

Mean Absolute Error (MAE)
Measures average prediction error.

Root Mean Squared Error (RMSE)
Penalizes large prediction errors.

R^2 Score (Coefficient of Determination)
Indicates how well the model explains variance in data.

Results (Example)

- MAE: Low error indicates accurate prediction.
- RMSE: Shows acceptable deviation from true values.
- R^2 Score: High value shows good model performance.

The predicted vs actual plot confirmed that the model captures general price trends effectively.

7. Model Saving and Deployment

The trained model was saved using a **pickle file**:

linear_regression_model.pkl

A small user interface was also created using Streamlit to allow users to enter housing features and predict house prices interactively.

8. Improvement Ideas

Although Linear Regression performs well, performance can be further improved by:

- Feature scaling and normalization
- Polynomial Regression
- Regularization techniques (Ridge/Lasso)
- Using advanced algorithms:

    - Random Forest Regression
    - Gradient Boosting
    - Neural Networks
- Hyperparameter tuning and cross-validation

9. Conclusion

This project successfully demonstrated the end-to-end machine learning lifecycle:

- Data Loading
- Exploratory Data Analysis
- Model Training
- Evaluation
- Model Saving and Prediction

The Linear Regression model effectively predicts California house prices and serves as a strong foundation for more advanced machine learning projects. This project is suitable for academic submission and portfolio presentation.